

DBSEDBD_TRIGGERS_AND_PROCEDURES

MANASVI GHAG

2520030335

Query 1: -CREATE DATABASE

```
1 • CREATE DATABASE BankDB;  
2 • USE BankDB;
```

Output :

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB

Query 2: - CREATE CUSTOMER TABLE

```
3 • CREATE TABLE Customer (  
4     Customer_ID INT PRIMARY KEY,  
5     Customer_Name VARCHAR(100) NOT NULL,  
6     Phone VARCHAR(15),  
7     Email VARCHAR(100),  
8     City VARCHAR(50)  
9 );
```

Output :

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ...

Query 3: - CREATE ACCOUNT TABLE

```
11 • CREATE TABLE Account (  
12     Account_No INT PRIMARY KEY,  
13     Customer_ID INT,  
14     Account_Type VARCHAR(20),  
15     Balance DECIMAL(12,2) DEFAULT 0,  
16     Branch VARCHAR(50),  
17  
18     FOREIGN KEY (Customer_ID)  
19     REFERENCES Customer(Customer_ID)  
20 );
```

Output

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ..
✓ 4	19:57:10	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ..

Query 4: - CREATE TRANSACTION TABLE

```
22 • CREATE TABLE Bank_Transaction (  
23     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
24     Account_No INT,  
25     Transaction_Type VARCHAR(20),  
26     Amount DECIMAL(12,2),  
27     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
28  
29     FOREIGN KEY (Account_No)  
30     REFERENCES Account(Account_No)  
31 );
```

Output

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ...
✓ 4	19:57:10	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...
✓ 5	19:58:05	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_I...

Query 5: - CREATE LOAN TABLE

```
33 • CREATE TABLE Loan (  
34     Loan_ID INT PRIMARY KEY,  
35     Customer_ID INT,  
36     Loan_Type VARCHAR(30),  
37     Loan_Amount DECIMAL(12,2),  
38     Interest_Rate DECIMAL(5,2),  
39  
40     FOREIGN KEY (Customer_ID)  
41     REFERENCES Customer(Customer_ID)  
42 );
```

Output

Action Output

#	Time	Action
✓ 1	19:55:43	CREATE DATABASE BankDB
✓ 2	19:55:44	USE BankDB
✓ 3	19:56:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name ...
✓ 4	19:57:10	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, ...
✓ 5	19:58:05	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_I...
✓ 6	19:59:20	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan...

Query 6: - INSERT CUSTOMER DATA

```
44 • INSERT INTO Customer  
45 (Customer_ID, Customer_Name, Phone, Email, City)  
46 VALUES  
47 (101, 'Manasvi', '9876543210', 'manasvi@gmail.com', 'Hyderabad'),  
48  
49 (102, 'Akhila', '9876543211', 'akhila@gmail.com', 'Vijayawada'),  
50  
51 (103, 'Hasini', '9876543212', 'hasini@gmail.com', 'Bangalore'),  
52  
53 (104, 'Bhavana', '9876543213', 'bhavana@gmail.com', 'Chennai'),  
54  
55 (105, 'Aswitha', '9876543214', 'aswitha@gmail.com', 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message
✓ 20	20:04:14	USE BankDB	0 row(s) affected
✓ 21	20:04:16	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Nam...	0 row(s) affected
✓ 22	20:04:19	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...	0 row(s) affected
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...	0 row(s) affected
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...	0 row(s) affected
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...	5 row(s) affected

Query 7: - INSERT ACCOUNT DATA

```
57 • INSERT INTO Account
58 (Account_No, Customer_ID, Account_Type, Balance, Branch)
59 VALUES
60 (10001, 101, 'Savings', 50000, 'Hyderabad'),
61
62 (10002, 102, 'Savings', 75000, 'Vijayawada'),
63
64 (10003, 103, 'Current', 120000, 'Bangalore'),
65
66 (10004, 104, 'Savings', 45000, 'Chennai'),
67
68 (10005, 105, 'Current', 90000, 'Hyderabad');
```

Output

Action Output

#	Time	Action
✓ 21	20:04:16	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Nam...
✓ 22	20:04:19	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✗ 26	20:05:11	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 27	20:06:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...

Query 8:- INSERT TRANSACTION DATA

```
70 • INSERT INTO Bank_Transaction
71 (Account_No, Transaction_Type, Amount)
72 VALUES
73 (10001, 'DEPOSIT', 10000),
74
75 (10002, 'DEPOSIT', 15000),
76
77 (10003, 'WITHDRAW', 20000),
78
79 (10004, 'DEPOSIT', 5000),
80
81 (10005, 'WITHDRAW', 10000);
```

Output

Action Output

#	Time	Action
✓ 22	20:04:19	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID IN...
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✗ 26	20:05:11	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 27	20:06:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...
✓ 28	20:07:06	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...

Query 9: - INSERT LOAN DATA

```
83 • INSERT INTO Loan
84 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
85 VALUES
86 (501, 101, 'Home Loan', 5000000, 7.5),
87
88 (502, 102, 'Education Loan', 1000000, 6.5),
89
90 (503, 103, 'Car Loan', 800000, 8.2),
91
92 (504, 104, 'Personal Loan', 500000, 10.5);
```

Output

Action Output

#	Time	Action
✓ 23	20:04:24	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_...
✓ 24	20:04:27	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Lo...
✓ 25	20:04:53	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✗ 26	20:05:11	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUE...
✓ 27	20:06:22	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branc...
✓ 28	20:07:06	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUE...
✓ 29	20:08:55	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_R...

Query 10: - DISPLAY TABLE DATA

```
94 • SELECT * FROM Customer;
```

```
95
```

Result Grid



Filter Rows:

Edit:



Export/Import

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Manasvi	9876543210	manasvi@gmail.com	Hyderabad
	102	Akhila	9876543211	akhila@gmail.com	Vijayawada
	103	Hasini	9876543212	hasini@gmail.com	Bangalore
	104	Bhavana	9876543213	bhavana@gmail.com	Chennai
	105	Aswitha	9876543214	aswitha@gmail.com	Hyderabad
✱	NULL	NULL	NULL	NULL	NULL

96 • `SELECT * FROM Account;`

97

Result Grid			Filter Rows:		Edit:			
	Account_No	Customer_ID	Account_Type	Balance	Branch			
▶	10001	101	Savings	50000.00	Hyderabad			
	10002	102	Savings	75000.00	Vijayawada			
	10003	103	Current	120000.00	Bangalore			
	10004	104	Savings	45000.00	Chennai			
	10005	105	Current	90000.00	Hyderabad			
✱	NULL	NULL	NULL	NULL	NULL			

98 • `SELECT * FROM Bank_Transaction;`

99

Result Grid			Filter Rows:		Edit:				Export/Import:
	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date				
▶	1	10001	DEPOSIT	10000.00	2026-08-16 20:07:06				
	2	10002	DEPOSIT	15000.00	2026-08-16 20:07:06				
	3	10003	WITHDRAW	20000.00	2026-08-16 20:07:06				
	4	10004	DEPOSIT	5000.00	2026-08-16 20:07:06				
	5	10005	WITHDRAW	10000.00	2026-08-16 20:07:06				
✱	NULL	NULL	NULL	NULL	NULL				

100 • `SELECT * FROM Loan;`

Result Grid			Filter Rows:		Edit:				Ex
	Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate				
▶	501	101	Home Loan	5000000.00	7.50				
	502	102	Education Loan	1000000.00	6.50				
	503	103	Car Loan	800000.00	8.20				
	504	104	Personal Loan	500000.00	10.50				
✱	NULL	NULL	NULL	NULL	NULL				

Query 11: - PROCEDURE TO FIND CUSTOMER ACCOUNTS

```
102 • CREATE PROCEDURE GetAllCustomers()
103 BEGIN
104     SELECT * FROM Customer;
105 END //
106 DELIMITER ;
107 • CALL GetAllCustomers();
108
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Manasvi	9876543210	manasvi@gmail.com	Hyderabad
	102	Akhila	9876543211	akhila@gmail.com	Vijayawada
	103	Hasini	9876543212	hasini@gmail.com	Bangalore
	104	Bhavana	9876543213	bhavana@gmail.com	Chennai
	105	Aswitha	9876543214	aswitha@gmail.com	Hyderabad

Query 12: - PROCEDURE TO DEPOSIT MONEY

```
109 DELIMITER //
110
111 • CREATE PROCEDURE DepositMoney(
112     IN p_Account_No INT,
113     IN p_Amount DECIMAL(12,2)
114 )
115 BEGIN
116     UPDATE Account
117     SET Balance = Balance + p_Amount
118     WHERE Account_No = p_Account_No;
119 END //
120 DELIMITER ;
121 • CALL DepositMoney(10001, 5000);
122
```

Output

Action Output

#	Time	Action
✗ 34	20:12:05	CALL GetAllCustomers()
✓ 35	20:12:13	SELECT * FROM Loan LIMIT 0, 1000
✗ 36	20:12:21	CALL GetAllCustomers()
✓ 37	20:12:26	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; E..
✓ 38	20:12:26	CALL GetAllCustomers()
✓ 39	20:14:17	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount ..
✓ 40	20:14:18	CALL DepositMoney(10001, 5000)

Query 13: - PROCEDURE TO WITHDRAW MONEY

```
124 DELIMITER //
125 • CREATE PROCEDURE WithdrawMoney(
126     IN p_Account_No INT,
127     IN p_Amount DECIMAL(12,2)
128 )
129 BEGIN
130     UPDATE Account
131     SET Balance = Balance - p_Amount
132     WHERE Account_No = p_Account_No;
133 END //
134 DELIMITER ;
135 • CALL WithdrawMoney(10001, 3000);
```

Output :

Action Output

#	Time	Action
✗ 36	20:12:21	CALL GetAllCustomers()
✓ 37	20:12:26	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; E...
✓ 38	20:12:26	CALL GetAllCustomers()
✓ 39	20:14:17	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount ...
✓ 40	20:14:18	CALL DepositMoney(10001, 5000)
✓ 41	20:15:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amou...
✓ 42	20:15:26	CALL WithdrawMoney(10001, 3000)

Query 14: - TRIGGER – PREVENT NEGATIVE BALANCE

```
137 DELIMITER //
138 • CREATE TRIGGER CheckBalance
139 BEFORE UPDATE ON Account
140 FOR EACH ROW
141 BEGIN
142     IF NEW.Balance < 0 THEN
143         SIGNAL SQLSTATE '45000'
144         SET MESSAGE_TEXT =
145             'Transaction failed: Insufficient balance';
146     END IF;
147 END //
148 DELIMITER ;
149
```

Output :

Action Output

#	Time	Action
✓ 37	20:12:26	CREATE PROCEDURE GetAllCustomers() BEGIN SELECT * FROM Customer; E...
✓ 38	20:12:26	CALL GetAllCustomers()
✓ 39	20:14:17	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount ..
✓ 40	20:14:18	CALL DepositMoney(10001, 5000)
✓ 41	20:15:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amou...
✓ 42	20:15:26	CALL WithdrawMoney(10001, 3000)
✓ 43	20:16:57	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH R...

Query 15: - CREATE TRANSACTION AUDIT TABLE

```
150 • CREATE TABLE Transaction_Audit (  
151     Audit_ID INT PRIMARY KEY AUTO_INCREMENT,  
152     Transaction_ID INT,  
153     Account_No INT,  
154     Transaction_Type VARCHAR(20),  
155     Amount DECIMAL(12,2),  
156     Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP  
157 );  
158  
159
```

Output

Action Output

#	Time	Action
✓ 38	20:12:26	CALL GetAllCustomers()
✓ 39	20:14:17	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount ...
✓ 40	20:14:18	CALL DepositMoney(10001, 5000)
✓ 41	20:15:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amou...
✓ 42	20:15:26	CALL WithdrawMoney(10001, 3000)
✓ 43	20:16:57	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH R...
✓ 44	20:19:01	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCR...

Query 16: - TRIGGER – TRANSACTION AUDIT

```
165 (   
166     Transaction_ID,  
167     Account_No,  
168     Transaction_Type,  
169     Amount  
170 )  
171 VALUES  
172 (   
173     NEW.Transaction_ID,  
174     NEW.Account_No,  
175     NEW.Transaction_Type,  
176     NEW.Amount  
177 );  
178 END //  
179 DELIMITER ;  
180
```

Output

Action Output

#	Time	Action
✓ 39	20:14:17	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount ..
✓ 40	20:14:18	CALL DepositMoney(10001, 5000)
✓ 41	20:15:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amou...
✓ 42	20:15:26	CALL WithdrawMoney(10001, 3000)
✓ 43	20:16:57	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH R...
✓ 44	20:19:01	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCR..
✓ 45	20:20:01	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR..

Query 17: - PROCEDURE – ACCOUNT TRANSFER

```
193 WHERE Account_No = SenderAccount;
194 IF SenderBalance < TransferAmount THEN
195     SIGNAL SQLSTATE '45000'
196     SET MESSAGE_TEXT =
197     'Transfer failed: Insufficient balance';
198 ELSE
199     UPDATE Account
200     SET Balance = Balance - TransferAmount
201     WHERE Account_No = SenderAccount;
202     UPDATE Account
203     SET Balance = Balance + TransferAmount
204     WHERE Account_No = ReceiverAccount;
205 END IF;
206 END //
207 DELIMITER ;
208 • CALL TransferMoney(10001, 10002, 5000);
```

Output

Action Output

#	Time	Action
✓ 41	20:15:26	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amou...
✓ 42	20:15:26	CALL WithdrawMoney(10001, 3000)
✓ 43	20:16:57	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH R...
✓ 44	20:19:01	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCR...
✓ 45	20:20:01	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR...
✓ 46	20:22:32	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN Receiver...
✓ 47	20:22:32	CALL TransferMoney(10001, 10002, 5000)

CHECK:

```
210 • SELECT *
211 FROM Account
212 WHERE Account_No IN (10001,10002);
213
```

Result Grid



Filter Rows:

Edit:



	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	47000.00	Hyderabad
	10002	102	Savings	80000.00	Vijayawada
*	NULL	NULL	NULL	NULL	NULL

Query 18: - PROCEDURE WITH IN, OUT PARAMETERS

The screenshot shows a database query editor with the following SQL code:

```
222 FROM Account
223 WHERE Account_No = p_Account_No;
224 END //
225 DELIMITER ;
226 • CALL GetBalance(10001, @CurrentBalance);
227 • SELECT @CurrentBalance;
228
```

Below the code, the 'Result Grid' shows the output of the query:

	@CurrentBalance
▶	47000.00

At the bottom, the 'Output' pane shows the 'Action Output' log:

#	Time	Action
✓ 45	20:20:01	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR...
✓ 46	20:22:32	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN Receiver...
✓ 47	20:22:32	CALL TransferMoney(10001, 10002, 5000)
✓ 48	20:23:21	SELECT * FROM Account WHERE Account_No IN (10001,10002) LIMIT 0, 1000
✓ 49	20:25:28	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance...
✓ 50	20:25:32	CALL GetBalance(10001, @CurrentBalance)
✓ 51	20:25:32	SELECT @CurrentBalance LIMIT 0, 1000

Query 19: -TRIGGER COMMANDS

The screenshot shows a database query editor with the following SQL code:

```
229 • SHOW TRIGGERS;
230
```

Below the code, the 'Result Grid' shows the output of the query:

	Trigger	Event	Table	Statement	Timing	Created
▶	CheckBalance	UPDATE	account	BEGIN IF NEW.Balance < 0 THEN SIG...	BEFORE	2026
	TransactionAudit	INSERT	bank_transaction	BEGIN INSERT INTO Transaction_Audit ...	AFTER	2026

Query 20: - PROCEDURE MANAGEMENT COMMANDS

```
229 • SHOW TRIGGERS;
230 • SHOW PROCEDURE STATUS
231 WHERE Db = 'BankDB';
232
```

Result Grid							
		Filter Rows:			Export:	Wrap Cell Content:	
	Db	Name	Type	Definer	Modified	Created	Se
►	bankdb	DepositMoney	PROCEDURE	root@localhost	2026-08-16 20:14:17	2026-08-16 20:14:17	DEF
	bankdb	GetAllCustomers	PROCEDURE	root@localhost	2026-08-16 20:12:26	2026-08-16 20:12:26	DEF
	bankdb	GetBalance	PROCEDURE	root@localhost	2026-08-16 20:25:28	2026-08-16 20:25:28	DEF
	bankdb	TransferMoney	PROCEDURE	root@localhost	2026-08-16 20:22:32	2026-08-16 20:22:32	DEF
	bankdb	WithdrawMoney	PROCEDURE	root@localhost	2026-08-16 20:15:26	2026-08-16 20:15:26	DEF